COURSE PROJECT TITLE IN ALL CAPS

A COURSE PROJECT REPORT

submitted by

STUDENT ONE (MUT24XX000)

STUDENT TWO (MUT24XX000)

STUDENT THREE (MUT24XX000)

STUDENT FOUR (MUT24XX000)

as part of the course

PBXXXX00 – COURSE NAME

to

The APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology

in

Artificial Intelligence and Data Science



Department of Artificial Intelligence and Data Science

Muthoot Institute of Technology and Science (MITS)

Varikoli, Ernakulam, Kerala

OCTOBER 2026

CERTIFICATE

This is to certify that the report entitled **“COURSE PROJECT TITLE IN ALL CAPS”**, submitted by **STUDENT ONE (MUT24XX000)**, **STUDENT TWO (MUT24XX000)**, **STUDENT THREE (MUT24XX000)**, and **STUDENT FOUR (MUT24XX000)** to **Muthoot Institute of Technology and Science (MITS)** for the award of the degree of **Bachelor of Technology** in **Artificial Intelligence and Data Science**, is a bonafide record of the course project work carried out by them, as part of the course **PBXXXX00 – COURSE NAME**, under my supervision and guidance.

Place: Varikoli
Date: 22/10/2026

Faculty-in-Charge
Department of Artificial Intelligence and Data Science

ABSTRACT

[Abstract text goes here – a concise, self-contained summary of the problem background, mathematical/computational methods, dataset scope, key simulation results, and practical engineering significance.]

CO – PO / PSO ARTICULATION MATRIX

Course Outcomes (COs) addressed by this project:

- **CO1:** Explain fundamental graph concepts and algorithms to analyse and solve problems.
- **CO2:** Illustrate matching and covering problems in graphs using algorithms and theorems.
- **CO3:** Apply connectivity and paths including Network Flow and Max-Flow Min-Cut Theorem.
- **CO4:** Illustrate Vertex Coloring, Upper Bounds, Brooks' Theorem, and Color-Critical Graphs.
- **CO5:** Apply concepts of planar graphs, Kuratowski's and Wagner's theorems, TSP, and dominating sets.

Table 1: Course Outcomes to Program Outcomes Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	-	-	-	-	-	-	-	-
CO2	3	3	3	-	-	-	-	-	-	-	-	-
CO3	3	3	3	3	-	-	-	-	-	-	-	-
CO4	3	3	3	3	-	-	-	-	-	-	-	-
CO5	3	3	3	-	-	-	-	-	-	-	-	-

SUSTAINABLE DEVELOPMENT GOALS (SDG) ADDRESSED

Table 2: Sustainable Development Goals Alignment

Goal	Target	Justification
SDG 11: Sustainable Cities	**Target 11.2:** Safe, sustainable transport systems.	[Project contribution to sustainable mobility and infrastructure optimization.]
SDG 9: Industry & Infrastructure	**Target 9.1:** Quality, reliable infrastructure.	[Project contribution to resilient infrastructure design and capital efficiency.]

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CHAPTER 1

INTRODUCTION

1.1 Background and Context

[Provide introductory background on the domain and application.]

1.2 Motivation

[Describe the motivation for employing computational graph algorithms.]

1.3 Scope of the Project

[Delineate the explicit engineering scope.]

1.4 Organization of the Report

[Brief chapter-by-chapter outline of the report.]

CHAPTER 2

PROBLEM DEFINITION AND OBJECTIVES

2.1 Problem Statement

[State the exact mathematical/algorithmic problem statement.]

2.2 Objectives

- Objective 1
- Objective 2
- Objective 3
- Objective 4

CHAPTER 3

DATASET AND NETWORK PREPROCESSING

3.1 Data Acquisition

[Describe source data extraction and preprocessing pipeline.]

3.2 Graph Model Formulation

[Define the vertex and directed edge attributes.]

CHAPTER 4

EXPLORATORY GRAPH ANALYSIS

4.1 Topological Properties

[Examine degree distributions, sparsity, clustering, and connectivity.]

CHAPTER 5

ALGORITHMIC METHODOLOGY

5.1 Core Algorithms

[Detail mathematical formulations, equilibrium conditions, and search heuristics.]

CHAPTER 6

EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

6.1 Evaluation Metrics

[Define quantitative performance indicators.]

6.2 Comparative Experimental Results

[Present experimental results across baseline and surge traffic scenarios.]

CHAPTER 7

SYSTEM ARCHITECTURE AND IMPLEMENTATION

7.1 Architecture Overview

[Detail system components, backend APIs, and frontend GIS interface.]

CHAPTER 8

DISCUSSION AND SENSITIVITY ANALYSIS

8.1 Engineering Insights

[Interpret systemic implications, physical tradeoffs, and computational scaling.]

CHAPTER 9

LIMITATIONS AND FUTURE RESEARCH

9.1 Limitations

[State theoretical or empirical boundary conditions.]

9.2 Future Directions

[Outline future extensions and research avenues.]

CHAPTER 10

CONCLUSION

[Summarize project achievements, concrete findings, and engineering takeaways.]

BIBLIOGRAPHY

[1] Author Name, “Paper or Book Title,” *Journal / Publisher*, vol. 1, no. 1, pp. 1–10, 2026.